

History of AI: 30 Key Figures

Evolution of Artificial Intelligence

From Turing (1950) to the ChatGPT Era (2022–)

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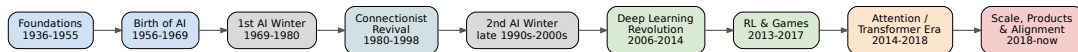
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July 13, 2026

- 1 Foundations (1936–1955)
- 2 Birth of AI (1956–1969)
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- 8 The Modern Era: Games, Science, and ChatGPT (2016–now)
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The Big Picture: 9 Eras of AI History



From Turing's question "Can machines think?" (1950) to today's large language models — with two "AI Winters" along the way.

Era 1: Foundations

1936–1955

Can machines think? Cybernetics, information theory

1. Alan Turing (1912–1954)



Alan Turing, British mathematician & codebreaker

Key work / paper: *Computing Machinery and Intelligence, Mind*, 1950
— proposed the **Turing Test** (Imitation Game)

- Reframed “can machines think?” as an empirical, behavioral test rather than a philosophical puzzle
- Also father of the Turing Machine (1936) — theoretical foundation of computing itself
- Every debate about machine intelligence (chatbots to LLMs) still traces back to this paper

Wikipedia: en.wikipedia.org/wiki/Alan_Turing Paper: [doi:10.1093/mind/LIX.236.433](https://doi.org/10.1093/mind/LIX.236.433)

1. Alan Turing (1912–1954) — Key Work / Event

A. M. Turing (1950) *Computing Machinery and Intelligence*. *Mind* 49: 433–460.

COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

1. The Imitation Game

I propose to consider the question, "Can machines think?" This should begin with definitions of the meaning of the terms "machine" and "think." The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words "machine" and "think" are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, "Can machines think?" is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

The new form of the problem can be described in terms of a game which we call the 'imitation game.' It is played with three people, a man (A), a woman (B), and an interrogator (C) who may be of either sex. The interrogator stays in a room apart from the other two. The object of the game for the interrogator is to determine which of the other two is the man and which is the woman. He knows them by labels X and Y, and at the end of the game he says either "X is A and Y is B" or "X is B and Y is A." The

"Computing Machinery and Intelligence," Mind, 1950

2. Norbert Wiener (1894–1964)



Norbert Wiener, founder of cybernetics

Key work / paper: *Cybernetics: Or Control and Communication in the Animal and the Machine*, 1948

- Introduced feedback loops and self-regulating systems — conceptual ancestors of reinforcement learning & robotics
- Coined the field of **cybernetics**: control + communication in animals and machines
- One of the earliest to warn about automation displacing human labor

Wikipedia: en.wikipedia.org/wiki/Norbert_Wiener

2. Norbert Wiener (1894–1964) — Key Work / Event

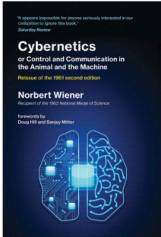
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Cybernetics or Control and Communication in the Animal and the Machine

By Norbert Wiener

The MIT Press

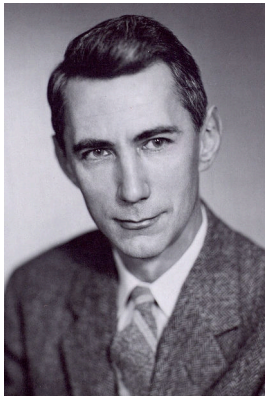
DOI: <https://doi.org/10.7551/mitpress/11810.001.0001>

ISBN electronic: 9780262355902

Publication date: 2019

Cybernetics, Reissue of the 1961 2nd Edition (book cover)

3. Claude Shannon (1916–2001)



Claude Shannon, father of information theory

Key work / paper: *A Mathematical Theory of Communication*, 1948;
co-organizer of 1956 Dartmouth Conference

- Information theory (bits, entropy) underlies neural net loss functions and LLM tokenization
- Built early theoretical framework for chess-playing programs
- Co-signed the 1955 Dartmouth proposal — one of AI's founding fathers

Wikipedia: en.wikipedia.org/wiki/Claude_Shannon

3. Claude Shannon (1916–2001) — Key Work / Event

A Mathematical Theory of Communication

By C. E. SHANNON

INTRODUCTION

THE recent development of various methods of modulation such as PCM and PPM which exchange bandwidth for signal-to-noise ratio has intensified the interest in a general theory of communication. A basis for such a theory is contained in the important papers of Nyquist¹ and Hartley² on this subject. In the present paper we will extend the theory to include a number of new factors, in particular the effect of noise in the channel, and the savings possible due to the statistical structure of the original message and due to the nature of the final destination of the information.

The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point. Frequently the messages have *meaning*; that is they refer to or are correlated according to some system with certain physical or conceptual entities. These semantic aspects of communication are irrelevant to the engineering problem. The significant aspect is that the actual message is one *selected from a set* of possible messages. The system must be designed to operate for each possible selection, not just the one which will actually be chosen since this is unknown at the time of design.

If the number of messages in the set is finite then this number or any monotonic function of this number can be regarded as a measure of the information produced when one message is chosen from the set, all choices being equally likely. As was pointed out by Hartley the most natural

A Mathematical Theory of Communication, 1948

Era 2: The Birth of AI

1956–1969

Dartmouth Conference, symbolic AI, first chatbot

4. John McCarthy (1927–2011)



John McCarthy, coined the term "AI"

Key work / paper: 1956 Dartmouth Summer Research Project on Artificial Intelligence (organizer)

- Coined the term **"Artificial Intelligence"** in the 1955 Dartmouth funding proposal
- Invented **LISP** (1958) — dominant AI programming language for decades
- The Dartmouth Workshop (summer 1956) is universally treated as AI's "birth date"

Wikipedia: [en.wikipedia.org/wiki/John_McCarthy_\(computer_scientist\)](https://en.wikipedia.org/wiki/John_McCarthy_(computer_scientist))

A PROPOSAL FOR THE
DARTMOUTH SUMMER RESEARCH PROJECT
ON ARTIFICIAL INTELLIGENCE

1955 Dartmouth funding proposal (first use of “artificial intelligence”)

5. Marvin Minsky (1927–2016)



Marvin Minsky, co-founder MIT AI Lab

Key work / paper: *Perceptrons*, 1969, with Seymour Papert

- Mathematically proved single-layer perceptrons cannot solve XOR / linearly-inseparable problems
- Widely credited with triggering the **1st AI Winter** for neural networks
- Redirected the field toward symbolic/rule-based AI for over a decade

Wikipedia: en.wikipedia.org/wiki/Marvin_Minsky

5. Marvin Minsky (1927–2016) — Key Work / Event

Perceptrons (book)

🌐 2 languages

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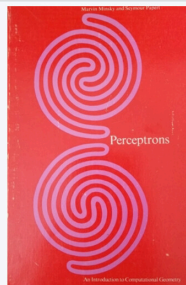
From Wikipedia, the free encyclopedia

Perceptrons: An Introduction to Computational Geometry is a book written by [Marvin Minsky](#) and [Seymour Papert](#) and published in 1969. An edition with handwritten corrections and additions was released in the early 1970s. An expanded edition was further published in 1988 (ISBN 9780262631112) after the [revival of neural networks](#), containing a chapter dedicated to countering the criticisms made of it in the 1980s.

The main subject of the book is the [perceptron](#), a type of [artificial neural network](#) developed in the late 1950s and early 1960s. The book was dedicated to psychologist [Frank Rosenblatt](#), who in 1957 had published the first model of a "Perceptron".^[1] Rosenblatt and Minsky knew each other since adolescence, having studied with a one-year difference at the [Bronx High School of Science](#).^[2] They became at one point central figures of a debate inside the AI research community, and are known to have promoted loud discussions in conferences, yet remained friendly.^[3]

This book is the center of a long-standing controversy in the study of [artificial intelligence](#). It is claimed that pessimistic predictions made by the authors were responsible for a change in the direction of research in AI, concentrating efforts on so-called "symbolic" systems, a line of research that petered out and contributed to the so-called [AI winter](#) of the 1980s, when AI's promise was

Perceptrons: An Introduction to Computational Geometry



Author [Marvin Minsky](#), [Seymour Papert](#)
Publication date 1969

Perceptrons, 1969 (book cover, with Seymour Papert)

6. Allen Newell (1927–1992)



Allen Newell, Carnegie Mellon

Key work / paper: Logic Theorist (1956) / General Problem Solver (1957)

- Co-created the **Logic Theorist** (1956) — arguably the first AI program, proved math theorems
- Co-created the **General Problem Solver (GPS, 1957)** — an early universal reasoning engine
- Established the “physical symbol system hypothesis”: intelligence = symbol manipulation

Wikipedia: en.wikipedia.org/wiki/Allen_Newell

Logic Theorist

🌟 9 languages ▾

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From Wikipedia, the free encyclopedia

Logic Theorist is a computer program completed in 1956 by [Allen Newell](#), [Herbert A. Simon](#), and [Cliff Shaw](#).^[1] It was the first program deliberately engineered to perform [automated reasoning](#), and has been described as "the first [artificial intelligence](#) program".^{[1][a]} Logic Theorist proved 38 of the first 52 theorems in chapter two of [Whitehead](#) and [Bertrand Russell](#)'s *Principia Mathematica*, and found new and shorter proofs for some of them.^[3]

History [edit]

In 1955, when Newell and Simon began to work on the Logic Theorist, the field of [artificial intelligence](#) did not yet exist; the term "artificial intelligence" would not be coined until the following summer.^[b]

Simon was a [political scientist](#) who had previously studied the way bureaucracies function as well as developing his theory of [bounded rationality](#) (for which he would later win the [Nobel Memorial Prize in Economic Sciences](#) in 1978). He believed the study of business organizations requires, like artificial intelligence, an insight into the nature of human problem solving and [decision making](#). Simon has stated that when consulting at [RAND Corporation](#) in the early 1950s, he saw a printer typing out a map, using ordinary letters and punctuation as symbols. This led him to think that a machine that could manipulate symbols could simulate decision making and possibly even the process of human thought.^{[5][6]}

Logic Theorist, 1956

7. Herbert A. Simon (1916–2001)



Herbert Simon, Carnegie Mellon; Nobel laureate

Key work / paper: General Problem Solver (1957), with Allen Newell

- With Newell: Logic Theorist (1956) and GPS (1957) — founding programs of symbolic AI
- This paradigm (GOF AI) dominated AI research through the 1980s expert-systems era
- Won the **1978 Nobel Prize in Economics** (bounded rationality) and 1975 Turing Award (with Newell)

Wikipedia: en.wikipedia.org/wiki/Herbert_A._Simon

7. Herbert A. Simon (1916–2001) — Key Work / Event

Key Milestones

1955 - Logic Theorist

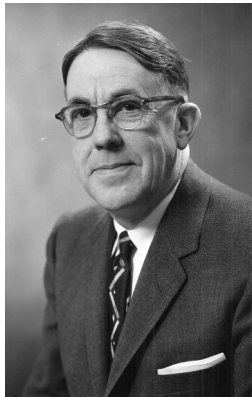
Developed by Allen Newell, Herbert A. Simon, and Cliff Shaw in 1955, the Logic Theorist is often considered the first artificial intelligence program. It was designed to mimic human problem-solving skills by proving mathematical theorems. The program used a method called heuristic search to solve problems, representing one of the first applications of heuristics in computing. The Logic Theorist was capable of proving many of the theorems in Whitehead and Russell's Principia Mathematica, and in some cases, it even found more elegant proofs than those originally published. Its success demonstrated the potential of machines to perform tasks that require intelligence, fundamentally challenging the prevailing views of the time about the capabilities of computers.

1957 - General Problem Solver

The General Problem Solver (GPS) was developed by Allen Newell and Herbert A. Simon in 1957 as an extension of the ideas introduced by the Logic Theorist. It was designed to be a more general problem-solving machine, capable of solving a wide range of problems, not just mathematical theorems. GPS attempted to model the general thought processes of humans when they face a problem, using a strategy that broke down problems into smaller, more manageable sub-problems. It applied rules and heuristics to solve these sub-problems, aiming to mimic human problem-solving techniques. Despite its limitations and the simplicity of the problems it could solve, the GPS was a groundbreaking step towards developing AI systems that could address complex tasks across different domains.

General Problem Solver, 1957

8. Arthur Samuel (1901–1990)



Arthur Samuel, IBM researcher

Key work / paper: Checkers-playing program (1959) that learned via self-play

- Wrote a checkers program that improved by playing against itself
- **Coined the term “machine learning”**
- Conceptual seed of self-play methods — direct ancestor of AlphaGo/AlphaZero

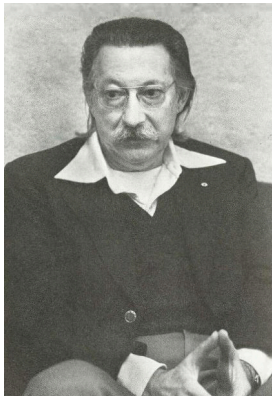
Wikipedia: en.wikipedia.org/wiki/Arthur_Samuel

8. Arthur Samuel (1901–1990) — Key Work / Event



Samuel's checkers program, 1959

9. Joseph Weizenbaum (1923–2008)



Joseph Weizenbaum, MIT

Key work / paper: *ELIZA – A Computer Program For the Study of Natural Language Communication...*, CACM, 1966

- Built **ELIZA** (1966), widely considered the **first chatbot** (a mock Rogerian psychotherapist)
- Discovered users attributed real understanding/empathy to it — the “**ELIZA effect**”
- Directly relevant today: why people anthropomorphize modern chatbots/LLMs

Wikipedia: en.wikipedia.org/wiki/ELIZA doi:10.1145/365153.365168

Computational Linguistics

A. G. OETTINGER, Editor

ELIZA—A Computer Program For the Study of Natural Language Communication Between Man And Machine

JOSEPH WEIZENBAUM

Massachusetts Institute of Technology, Cambridge, Mass.*

ELIZA is a program operating within the MAC time-sharing system at MIT which makes certain kinds of natural language conversation between man and computer possible. Input sentences are analyzed on the basis of decomposition rules which are triggered by key words appearing in the input text. Responses are generated by reassembly rules associated with selected decomposition rules. The fundamental technical problems with which ELIZA is concerned are: (1) the identification of key words, (2) the discovery of minimal context, (3) the choice of appropriate transformations, (4) generation of responses in the absence of key words, and (5) the provision of an editing capability for ELIZA "scripts". A discussion of some psychological issues relevant to the ELIZA approach as well as of future developments concludes the paper.

The object of this paper is to cause just such a re-evaluation of the program about to be "explained". Few programs ever needed it more.

ELIZA Program

ELIZA is a program which makes natural language conversation with a computer possible. Its present implementation is on the MAC time-sharing system at MIT. It is written in MAD-SLIP [4] for the IBM 7094. Its name was chosen to emphasize that it may be incrementally improved by its users, since its language abilities may be continually improved by a "teacher". Like the Eliza of Pygmalion fame, it can be made to appear even more civilized, the relation of appearance to reality, however, remaining in the domain of the playwright.

For the present purpose it is sufficient to characterize the MAC system as one which permits an individual to operate a full scale computer from a remotely located typewriter. The individual operator has the illusion that he is the sole user of the computer complex, while in fact others may be "time-sharing" the system with him. What is important here is that the computer can read messages typed on the typewriter and respond by writing on the same instrument. The time between the computer's receipt of a message and the appearance of its response is a function of the program controlling the dialogue and of

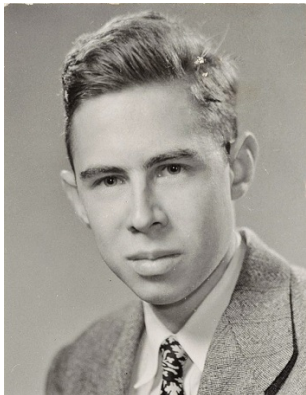
ELIZA paper

Era 3: Connectionist Revival

1980–1998

Neuro-inspired vision, backprop, and the road out of the 1st AI Winter

10. Frank Rosenblatt (1928–1971)



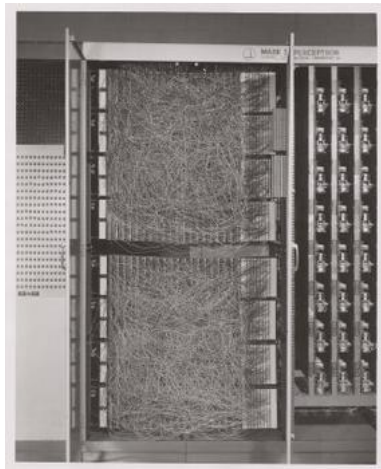
Frank Rosenblatt with the Mark I Perceptron

Key work / paper: *The Perceptron: A Probabilistic Model for Information Storage...*, Psychological Review, 1958

- Built the **Perceptron** — first trainable neural network machine (Mark I hardware)
- First demonstration that a machine could learn from labeled examples via adjustable weights
- Direct ancestor of every neuron in today's deep nets; limitations (Minsky/Papert, 1969) triggered the 1st AI Winter

Wikipedia: en.wikipedia.org/wiki/Perceptron

10. Frank Rosenblatt (1928–1971) — Key Work / Event



Mark I Perceptron machine, the first implementation of the perceptron algorithm.

11. Kunihiro Fukushima (b. 1936)



Kunihiro Fukushima, Japanese computer scientist

Key work / paper: *Neocognitron: A Self-Organizing Neural Network Model...*, Biological Cybernetics, 1980

- Built the **Neocognitron** — true architectural ancestor of the CNN, inspired by visual-cortex biology
- Introduced local receptive fields + hierarchical, shift-invariant feature detection
- The “missing link” between 1980s neuroscience-inspired nets and LeNet/AlexNet

Wikipedia: en.wikipedia.org/wiki/Kunihiro_Fukushima doi:10.1007/BF00344251

11. Kunihiro Fukushima (b. 1936) — Key Work / Event

Biol. Cybernetics 36, 193–202 (1980)

Biological
Cybernetics
© by Springer-Verlag 1980

Neocognitron: A Self-organizing Neural Network Model for a Mechanism of Pattern Recognition Unaffected by Shift in Position

Kunihiro Fukushima

NHK Broadcasting Science Research Laboratories, Kinuta, Setagaya, Tokyo, Japan

Abstract. A neural network model for a mechanism of visual pattern recognition is proposed in this paper. The network is self-organized by “learning without a teacher”, and acquires an ability to recognize stimulus patterns based on the geometrical similarity (Gestalt) of their shapes without affected by their positions. This network is given a nickname “neocognitron”. After completion of self-organization, the network has a structure similar to the hierarchy model of the visual nervous system proposed by Hubel and Wiesel. The network consists of an input layer (photoreceptor

reveal it only by conventional physiological experiments. So, we take a slightly different approach to this problem. If we could make a neural network model which has the same capability for pattern recognition as a human being, it would give us a powerful clue to the understanding of the neural mechanism in the brain. In this paper, we discuss how to synthesize a neural network model in order to endow it an ability of pattern recognition like a human being.

Several models were proposed with this intention (Rosenblatt, 1962; Kabrisky, 1966; Giebel, 1971;

Neocognitron architecture paper, 1980

12. David Rumelhart (1942–2011)



David Rumelhart, PDP group, UC San Diego

Key work / paper: *Learning representations by back-propagating errors*, Nature, 1986

- Co-authored the **backpropagation** paper (with Hinton & Williams) — made training multi-layer nets practical
- Arguably the single most important algorithmic paper in deep learning history
- Enables everything from 1990s handwriting recognition to today's trillion-parameter LLMs

Wikipedia: en.wikipedia.org/wiki/David_Rumelhart Nature 323, 533–536 (1986)

12. David Rumelhart (1942–2011) — Key Work / Event

Learning representations by back-propagating errors

David E. Rumelhart*, Geoffrey E. Hinton†
& Ronald J. Williams*

* Institute for Cognitive Science, C-015, University of California,
San Diego, La Jolla, California 92093, USA

† Department of Computer Science, Carnegie-Mellon University,
Pittsburgh, Philadelphia 15213, USA

We describe a new learning procedure, back-propagation, for networks of neurone-like units. The procedure repeatedly adjusts the weights of the connections in the network so as to minimize a measure of the difference between the actual output vector of the net and the desired output vector. As a result of the weight adjustments, internal 'hidden' units which are not part of the input or output come to represent important features of the task domain, and the regularities in the task are captured by the interactions of these units. The ability to create useful new features distinguishes back-propagation from earlier, simpler methods such as the perceptron-convergence procedure¹.

There have been many attempts to design self-organizing neural networks. The aim is to find a powerful synaptic modification rule that will allow an arbitrarily connected neural network to develop an internal structure that is appropriate for a particular task domain. The task is specified by giving the desired state vector of the output units for each state vector of the input units. If the input units are directly connected to the output units it is relatively easy to find learning rules that iteratively adjust the relative strengths of the connections so as to progressively reduce the difference between the actual and desired output vectors². Learning becomes more interesting but

more difficult when we introduce hidden units whose actual or desired states are not specified by the task. (In perceptrons, there are 'feature analysers' between the input and output that are not true hidden units because their input connections are fixed by hand, so their states are completely determined by the input vector: they do not learn representations.) The learning procedure must decide under what circumstances the hidden units should be active in order to help achieve the desired input-output behaviour. This amounts to deciding what these units should represent. We demonstrate that a general purpose and relatively simple procedure is powerful enough to construct appropriate internal representations.

The simplest form of the learning procedure is for layered networks which have a layer of input units at the bottom; any number of intermediate layers; and a layer of output units at the top. Connections within a layer or from higher to lower layers are forbidden, but connections can skip intermediate layers. An input vector is presented to the network by setting the states of the input units. Then the states of the units in each layer are determined by applying equations (1) and (2) to the connections coming from lower layers. All units within a layer have their states set in parallel, but different layers have their states set sequentially, starting at the bottom and working upwards until the states of the output units are determined.

The total input, x_j , to unit j is a linear function of the outputs, y_i , of the units that are connected to j and of the weights, w_{ji} , on these connections

$$x_j = \sum_i y_i w_{ji} \quad (1)$$

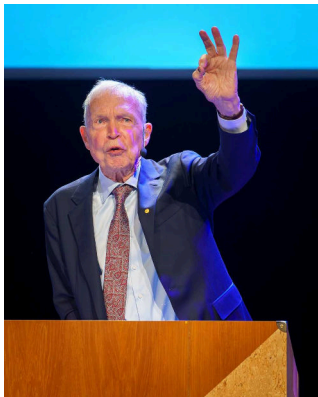
Units can be given biases by introducing an extra input to each unit which always has a value of 1. The weight on this extra input is called the bias and is equivalent to a threshold of the opposite sign. It can be treated just like the other weights.

A unit has a real-valued output, y_j , which is a non-linear function of its total input

1

Rumelhart, Hinton & Williams (1986), Nature, first page

13. John J. Hopfield (b. 1933)



John Hopfield, Princeton (emeritus)

Key work / paper: *Neural networks and physical systems with emergent collective computational abilities*, PNAS, 1982

- Built the **Hopfield Network** — associative memory using physics-based energy minimization
- Revitalized neural net research during the 1st AI Winter
- **2024 Nobel Prize in Physics** jointly with Geoffrey Hinton — first Nobel explicitly for AI/ML

Wikipedia: en.wikipedia.org/wiki/John_Hopfield

13. John J. Hopfield (b. 1933) — Key Work / Event

Proc. Natl. Acad. Sci. USA
Vol. 79, pp. 2554–2558, April 1982
Biophysics

Neural networks and physical systems with emergent collective computational abilities

(associative memory/parallel processing/categorization/content-addressable memory/fail-soft devices)

J. J. HOPFIELD

Division of Chemistry and Biology, California Institute of Technology, Pasadena, California 91125; and Bell Laboratories, Murray Hill, New Jersey 07974

Contributed by John J. Hopfield, January 15, 1982

ABSTRACT Computational properties of use to biological organisms or to the construction of computers can emerge as collective properties of systems having a large number of simple equivalent components (or neurons). The physical meaning of content-addressable memory is described by an appropriate phase space flow of the state of a system. A model of such a system is given, based on aspects of neurobiology but readily adapted to integrated circuits. The collective properties of this model produce a content-addressable memory which correctly yields an entire memory from any subpart of sufficient size. The algorithm for the time evolution of the state of the system is based on asynchronous

calized content-addressable memory or categorizer using extensive asynchronous parallel processing.

The general content-addressable memory of a physical system

Suppose that an item stored in memory is “H. A. Kramers & G. H. Wannier *Phys. Rev.* **60**, 252 (1941).” A general content-addressable memory would be capable of retrieving this entire memory item on the basis of sufficient partial information. The input “& Wannier, (1941)” might suffice. An ideal memory could deal with errors and retrieve this reference even from the

Hopfield network paper, 1982

14. Geoffrey E. Hinton (b. 1947)



Geoffrey Hinton, "Godfather of Deep Learning"

Key work / paper: Backpropagation (1986); Deep Belief Nets (2006); **AlexNet** (2012, with Krizhevsky & Sutskever)

- Bridges the 1980s connectionist revival and the 2012+ deep learning explosion
- **2018 Turing Award** (with Bengio, LeCun) and **2024 Nobel Prize in Physics** (with Hopfield)
- Left Google in 2023; now a prominent public voice on AI safety/existential risk

Wikipedia: en.wikipedia.org/wiki/Geoffrey_Hinton

ImageNet Classification with Deep Convolutional Neural Networks

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Abstract

We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet ILSVRC-2010 contest into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 37.5% and 17.0% which is considerably better than the previous state-of-the-art. The neural network, which has 60 million parameters and 650,000 neurons, consists of five convolutional layers, some of which are followed by max-pooling layers, and three fully-connected layers with a final 1000-way softmax. To make training faster, we used non-saturating neurons and a very efficient GPU implementation of the convolution operation. To reduce overfitting in the fully-connected layers we employed a recently-developed regularization method called “dropout” that proved to be very effective. We also entered a variant of this model in the ILSVRC-2012 competition and achieved a winning top-5 test error rate of 15.3%, compared to 26.2% achieved by the second-best entry.

AlexNet architecture paper, 2012

Era 4: The Deep Learning Trio

Computer vision (CNNs) and sequence models

15. Yann LeCun (b. 1960)



Yann LeCun, French-American computer scientist

Key work / paper: *Gradient-Based Learning Applied to Document Recognition*, 1998 (LeNet-5)

- Built the first practical **convolutional neural network (CNN)**, used commercially to read handwritten checks
- Established the CNN template (convolution + pooling + weight sharing) underlying modern computer vision
- **2018 Turing Award** (with Hinton and Bengio)

Wikipedia: en.wikipedia.org/wiki/Yann_LeCun *Homepage:* yann.lecun.com

Gradient-Based Learning Applied to Document Recognition

Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner

Abstract—

Multilayer Neural Networks trained with the backpropagation algorithm constitute the best example of a successful Gradient-Based Learning technique. Given an appropriate network architecture, Gradient-Based Learning algorithms can be used to synthesize a complex decision surface that can classify high-dimensional patterns such as handwritten characters, with minimal preprocessing. This paper reviews various methods applied to handwritten character recognition and compares them on a standard handwritten digit recognition task. Convolutional Neural Networks, that are specifically designed to deal with the variability of 2D shapes, are shown to outperform all other techniques.

Real-life document recognition systems are composed of multiple modules including field extraction, segmentation, recognition, and language modeling. A new learning paradigm, called Graph Transformer Networks (GTN), allows such multi-module systems to be trained globally using Gradient-Based methods so as to minimize an overall performance measure.

Two systems for on-line handwriting recognition are described. Experiments demonstrate the advantage of global training, and the flexibility of Graph Transformer Networks.

A Graph Transformer Network for reading bank check is also described. It uses Convolutional Neural Network char-

I. INTRODUCTION

Over the last several years, machine learning techniques, particularly when applied to neural networks, have played an increasingly important role in the design of pattern recognition systems. In fact, it could be argued that the availability of learning techniques has been a crucial factor in the recent success of pattern recognition applications such as continuous speech recognition and handwriting recognition.

The main message of this paper is that better pattern recognition systems can be built by relying more on automatic learning, and less on hand-designed heuristics. This is made possible by recent progress in machine learning and computer technology. Using character recognition as a case study, we show that hand-crafted feature extraction can be advantageously replaced by carefully designed learning machines that operate directly on pixel images. Using document understanding as a case study, we show that the traditional way of building recognition systems by

LeNet-5 architecture paper, 1998

16. Yoshua Bengio (b. 1964)



Yoshua Bengio, Univ. of Montréal / Mila

Key work / paper: *Neural Machine Translation by Jointly Learning to Align and Translate*, 2014

- Co-author of LeNet-5; pioneered neural probabilistic language modeling (2003)
- Introduced the **attention mechanism** to sequence models (2014) — direct ancestor of the Transformer
- **2018 Turing Award** (with Hinton, LeCun); now a leading voice on AI safety/governance

Wikipedia: en.wikipedia.org/wiki/Yoshua_Bengio Homepage: yoshuabengio.org

16. Yoshua Bengio (b. 1964) — Key Work / Event



Computer Science > Computation and Language

[Submitted on 1 Sep 2014 (v1), last revised 19 May 2016 (this version, v7)]

Neural Machine Translation by Jointly Learning to Align and Translate

 [CODE](#)  [Notebook](#)    

Dzmitry Bahdanau, Kyunghyun Cho, Yoshua Bengio

Neural machine translation is a recently proposed approach to machine translation. Unlike the traditional statistical machine translation, the neural machine translation aims at building a single neural network that can be jointly tuned to maximize the translation performance. The models proposed recently for neural machine translation often belong to a family of encoder-decoders and consists of an encoder that encodes a source sentence into a fixed-length vector from which a decoder generates a translation. In this paper, we conjecture that the use of a fixed-length vector is a bottleneck in improving the performance of this basic encoder-decoder architecture, and propose to extend this by allowing a model to automatically (soft-)search for parts of a source sentence that are relevant to predicting a target word, without having to form these parts as a hard segment explicitly. With this new approach, we achieve a translation performance comparable to the existing state-of-the-art phrase-based system on the task of English-to-French translation. Furthermore, qualitative analysis reveals that the (soft-)alignments found by the model agree well with our intuition.

Comments: Accepted at ICLR 2015 as oral presentation

Subjects: **Computation and Language (cs.CL)**; Machine Learning (cs.LG); Neural and Evolutionary Computing (cs.NE); Machine Learning (stat.ML)

Bahdanau, Cho & Bengio (2014) attention paper

17. Jürgen Schmidhuber (b. 1963)



Jürgen Schmidhuber, IDSIA, Switzerland

Key work / paper: *Long Short-Term Memory*, Neural Computation, 1997 (with Sepp Hochreiter)

- Built **LSTM** — gated recurrent architecture solving the vanishing-gradient problem for long sequences
- Dominant sequence-model architecture for nearly two decades (speech, translation, time series)
- Early foundational work on GAN-like adversarial training and self-supervised pretraining (early 1990s)

Wikipedia: en.wikipedia.org/wiki/J%C3%BCrgen_Schmidhuber doi:10.1162/neco.1997.9.8.1735

LONG SHORT-TERM MEMORY

NEURAL COMPUTATION 9(8):1735–1780, 1997

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Abstract

Learning to store information over extended time intervals via recurrent backpropagation takes a very long time, mostly due to insufficient, decaying error back flow. We briefly review Hochreiter's 1991 analysis of this problem, then address it by introducing a novel, efficient, gradient-based method called "Long Short-Term Memory" (LSTM). Truncating the gradient where this does not do harm, LSTM can learn to bridge minimal time lags in excess of 1000 discrete time steps by enforcing *constant* error flow through "constant error carousels" within special units. Multiplicative gate units learn to open and close access to the constant error flow. LSTM is local in space and time; its computational complexity per time step and weight is $O(1)$. Our experiments with artificial data involve local, distributed, real-valued, and noisy pattern representations. In comparisons with RTRL, BPTT, Recurrent Cascade-Correlation, Elman nets, and Neural Sequence Chunking, LSTM leads to many more successful runs, and learns much faster. LSTM also solves complex, artificial long time lag tasks that have never been solved by previous recurrent network algorithms.

LSTM paper, 1997

Era 5: Statistical & Probabilistic ML

The “pre-deep-learning” gold standard

18. Vladimir Vapnik (b. 1936)



Vladimir Vapnik, statistician

Key work / paper: *Support-Vector Networks*, 1995 (with Corinna Cortes)

- Co-invented the **Support Vector Machine (SVM)**
- Developed **VC theory** — mathematical foundation of statistical learning theory
- SVMs were the dominant, most reliable ML method for classification through the 1990s–2000s

Wikipedia: en.wikipedia.org/wiki/Vladimir_Vapnik

18. Vladimir Vapnik (b. 1936) — Key Work / Event

Machine Learning, 20, 273–297 (1995)

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Support-Vector Networks

CORINNA CORTES

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Editor: Lorenza Saitta

Abstract. The *support-vector network* is a new learning machine for two-group classification problems. The machine conceptually implements the following idea: input vectors are non-linearly mapped to a very high-dimension feature space. In this feature space a linear decision surface is constructed. Special properties of the decision surface ensures high generalization ability of the learning machine. The idea behind the support-vector network was previously implemented for the restricted case where the training data can be separated without errors. We here extend this result to non-separable training data.

High generalization ability of support-vector networks utilizing polynomial input transformations is demonstrated. We also compare the performance of the support-vector network to various classical learning algorithms that all took part in a benchmark study of Optical Character Recognition.

SVM paper

19. Judea Pearl (b. 1936)



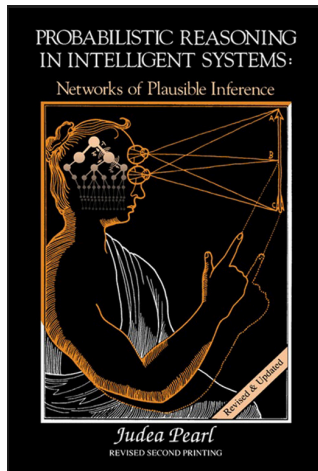
Judea Pearl, UCLA

Key work / paper: *Probabilistic Reasoning in Intelligent Systems*, 1988; *Causality*, 2000

- Gave AI a rigorous framework for reasoning under uncertainty: **Bayesian networks**
- Developed the **do-calculus** for causal inference (correlation vs. causation)
- **2011 Turing Award**; increasingly cited today to address LLM hallucination via causal reasoning

Wikipedia: en.wikipedia.org/wiki/Judea_Pearl

19. Judea Pearl (b. 1936) — Key Work / Event



Bayesian network book

Era 6: Deep Learning Revolution & Reinforcement Learning

2006–2017

GPUs + big data + deep nets; RL masters Atari, Go

20. Richard S. Sutton (b. 1957)



Richard Sutton, Univ. of Alberta / Amii

Key work / paper: *Reinforcement Learning: An Introduction*, 1998/2018 (with Andrew Barto)

- Co-developed **temporal difference (TD) learning** — core algorithmic foundation of RL
- Textbook cited 75,000+ times; RL underlies AlphaGo, robotics, and **RLHF** (used to align ChatGPT)
- **2024 ACM A.M. Turing Award** jointly with Andrew Barto

Wikipedia: en.wikipedia.org/wiki/Richard_S._Sutton

Reinforcement Learning: An Introduction

Second edition, in progress

Richard S. Sutton and Andrew G. Barto
© 2014, 2015

Reinforcement Learning: An Introduction, book cover

21. Andrew G. Barto (b. 1948)



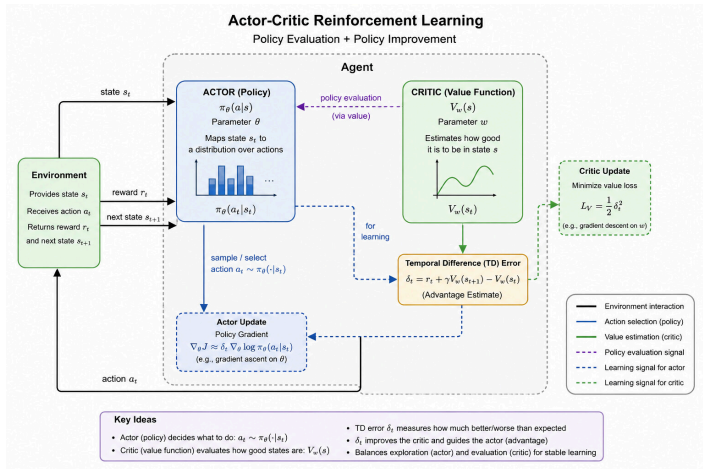
Andrew Barto, Univ. of Massachusetts Amherst (emeritus)

Key work / paper: Actor-critic methods, temporal difference learning (with Richard Sutton)

- Sutton's lifelong co-author; together built the algorithmic foundations of **reinforcement learning**
- This lineage runs through AlphaGo/AlphaZero's self-play and RLHF (GPT-3 → ChatGPT)
- **2024 ACM A.M. Turing Award**, jointly with Sutton — best presented as one “RL founders” story

Wikipedia: en.wikipedia.org/wiki/Andrew_Barto awards.acm.org/about/2024-turing

21. Andrew G. Barto (b. 1948) — Key Work / Event



Actor-critic RL diagram

22. Andrew Ng (b. 1976)



Andrew Ng, Stanford / Coursera / Google Brain

Key work / paper: Google Brain project (2011–2012): large-scale unsupervised learning from YouTube video

- Proved that scaling compute + data to neural nets yields qualitatively new capabilities (“cat detector”)
- A precursor argument to today’s “scaling laws”
- Democratized AI education worldwide via Coursera / Deep Learning Specialization (millions trained)

Wikipedia: en.wikipedia.org/wiki/Andrew_Ng Homepage: andrewng.org

22. Andrew Ng (b. 1976) — Key Work / Event



Computer Science > Machine Learning

[Submitted on 29 Dec 2011 (v1), last revised 12 Jul 2012 (this version, v5)]

Building high-level features using large scale unsupervised learning

 CODE  Notebook    

Quoc V. Le, Marc'Aurelio Ranzato, Rajat Monga, Matthieu Devin, Kai Chen, Greg S. Corrado, Jeff Dean, Andrew Y. Ng

 Ask

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We consider the problem of building high-level, class-specific feature detectors from only unlabeled data. For example, is it possible to learn a face detector using only unlabeled images? To answer this, we train a 9-layered locally connected sparse autoencoder with pooling and local contrast normalization on a large dataset of images (the model has 1 billion connections, the dataset has 10 million 200x200 pixel images downloaded from the Internet). We train this network using model parallelism and asynchronous SGD on a cluster with 1,000 machines (16,000 cores) for three days. Contrary to what appears to be a widely-held intuition, our experimental results reveal that it is possible to train a face detector without having to label images as containing a face or not. Control experiments show that this feature detector is robust not only to translation but also to scaling and out-of-plane rotation. We also find that the same network is sensitive to other high-level concepts such as cat faces and human bodies. Starting with these learned features, we trained our network to obtain 15.8% accuracy in recognizing 20,000 object categories from ImageNet, a leap of 70% relative improvement over the previous state-of-the-art.

Paper, 2012

23. Fei-Fei Li (b. 1976)



Fei-Fei Li, Stanford HAI

Key work / paper: *ImageNet: A Large-Scale Hierarchical Image Database*, 2009

- Built **ImageNet**: 14M+ labeled images, 22,000 categories; launched the ILSVRC competition
- “Data, not just algorithms” was the missing ingredient — enabled the 2012 AlexNet breakthrough
- Co-founded AI4ALL (diversity in AI); now building “spatial intelligence” at World Labs

Wikipedia: en.wikipedia.org/wiki/Fei-Fei_Li Stanford HAI bio: hai.stanford.edu

ImageNet: A Large-Scale Hierarchical Image Database

Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li and Li Fei-Fei
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{jiadeng, wdong, rsocher, jial, li, feifeili}@cs.princeton.edu

Abstract

The explosion of image data on the Internet has the potential to foster more sophisticated and robust models and algorithms to index, retrieve, organize and interact with images and multimedia data. But exactly how such data can be harnessed and organized remains a critical problem. We introduce here a new database called “ImageNet”, a large-scale ontology of images built upon the backbone of the WordNet structure. ImageNet aims to populate the majority of the 80,000 synsets of WordNet with an average of 500-1000 clean and full resolution images. This will result in tens of millions of annotated images organized by the semantic hierarchy of WordNet. This paper offers a detailed analysis of ImageNet in its current state: 12 subtrees with 5247 synsets and 3.2 million images in total. We show that ImageNet is much larger in scale and diversity and much more accurate than the current image datasets. Constructing such a large-scale database is a challenging task. We describe the data collection scheme with Amazon Mechanical Turk. Lastly, we illustrate the usefulness of ImageNet through three simple applications in object recognition, image classification and automatic object clustering. We hope

content-based image search and image understanding algorithms, as well as for providing critical training and benchmarking data for such algorithms.

ImageNet uses the hierarchical structure of WordNet [9]. Each meaningful concept in WordNet, possibly described by multiple words or word phrases, is called a “synonym set” or “synset”. There are around 80,000 noun synsets in WordNet. In ImageNet, we aim to provide on average 500-1000 images to illustrate each synset. Images of each concept are quality-controlled and human-annotated as described in Sec. 3.2. ImageNet, therefore, will offer tens of millions of cleanly sorted images. In this paper, we report the current version of ImageNet, consisting of 12 “subtrees”: *mammal, bird, fish, reptile, amphibian, vehicle, furniture, musical instrument, geological formation, tool, flower, fruit*. These subtrees contain 5247 synsets and 3.2 million images. Fig. 1 shows a snapshot of two branches of the mammal and vehicle subtrees. The database is publicly available at <http://www.image-net.org>.

The rest of the paper is organized as follows: We first show that ImageNet is a large-scale, accurate and diverse image database (Section 2). In Section 4, we present a few simple application examples by exploiting the current Im-

ImageNet paper, 2009

24. Ian Goodfellow (b. 1985)



Ian Goodfellow, formerly Google Brain / DeepMind / Apple

Key work / paper: *Generative Adversarial Networks*, 2014 (with Bengio et al.)

- Invented **GANs**: generator vs. discriminator trained in competition
- Launched the modern generative-AI-for-images era (deepfakes, style transfer, photorealistic synthesis)
- Preceded diffusion models (which took over image generation circa 2021–2022)

Wikipedia: en.wikipedia.org/wiki/Ian_Goodfellow arxiv.org/abs/1406.2661

24. Ian Goodfellow (b. 1985) — Key Work / Event



Statistics > Machine Learning

[Submitted on 10 Jun 2014]

Generative Adversarial Networks

 **CODE**  **Notebook**    

[Ian J. Goodfellow](#), [Jean Pouget-Abadie](#), [Mehdi Mirza](#), [Bing Xu](#), [David Warde-Farley](#), [Sherjil Ozair](#), [Aaron Courville](#), [Yoshua Bengio](#)

We propose a new framework for estimating generative models via an adversarial process, in which we simultaneously train two models: a generative model G that captures the data distribution, and a discriminative model D that estimates the probability that a sample came from the training data rather than G . The training procedure for G is to maximize the probability of D making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions G and D , a unique solution exists, with G recovering the training data distribution and D equal to $1/2$ everywhere. In the case where G and D are defined by multilayer perceptrons, the entire system can be trained with backpropagation. There is no need for any Markov chains or unrolled approximate inference networks during either training or generation of samples. Experiments demonstrate the potential of the framework through qualitative and quantitative evaluation of the generated samples.

GAN paper, 2014

25. Alex Krizhevsky (b. ~1986)



Alex Krizhevsky, PhD student of Hinton, Univ. of Toronto

Key work / paper: *ImageNet Classification with Deep Convolutional Neural Networks*, 2012 (with Sutskever & Hinton)

- Built **AlexNet** — won ImageNet 2012 by ~10.8 points over the runner-up
- Widely regarded as the actual “Big Bang” moment of the deep learning revolution
- Trained on 2 consumer GPUs; instantly ended the era of hand-crafted vision features / SVMs

Wikipedia: en.wikipedia.org/wiki/AlexNet

ImageNet Classification with Deep Convolutional Neural Networks

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Geoffrey E. Hinton
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Abstract

We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet ILSVRC-2010 contest into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 37.5% and 17.0% which is considerably better than the previous state-of-the-art. The neural network, which has 60 million parameters and 650,000 neurons, consists of five convolutional layers, some of which are followed by max-pooling layers, and three fully-connected layers with a final 1000-way softmax. To make training faster, we used non-saturating neurons and a very efficient GPU implementation of the convolution operation. To reduce overfitting in the fully-connected layers we employed a recently-developed regularization method called “dropout” that proved to be very effective. We also entered a variant of this model in the ILSVRC-2012 competition and achieved a winning top-5 test error rate of 15.3%, compared to 26.2% achieved by the second-best entry.

AlexNet paper, 2012

26. Ilya Sutskever (b. 1985)



Ilya Sutskever, co-founder & former Chief Scientist, OpenAI

Key work / paper: *Sequence to Sequence Learning with Neural Networks*, 2014

- Co-author of AlexNet (2012); established the encoder-decoder paradigm for sequence tasks
- Central technical architect behind GPT-1 through GPT-4 at OpenAI
- Now founder of Safe Superintelligence Inc., focused on AI safety before capability

Wikipedia: en.wikipedia.org/wiki/Ilya_Sutskever

26. Ilya Sutskever (b. 1985) — Key Work / Event



Computer Science > Computation and Language

[Submitted on 10 Sep 2014 (v1), last revised 14 Dec 2014 (this version, v3)]

Sequence to Sequence Learning with Neural Networks

 [CODE](#)  [Notebook](#)    

[Ilya Sutskever](#), [Oriol Vinyals](#), [Quoc V. Le](#)

 [Ask](#)

powered by [CatalyzeX](#)

Deep Neural Networks (DNNs) are powerful models that have achieved excellent performance on difficult learning tasks. Although DNNs work well whenever large labeled training sets are available, they cannot be used to map sequences to sequences. In this paper, we present a general end-to-end approach to sequence learning that makes minimal assumptions on the sequence structure. Our method uses a multilayered Long Short-Term Memory (LSTM) to map the input sequence to a vector of a fixed dimensionality, and then another deep LSTM to decode the target sequence from the vector. Our main result is that on an English to French translation task from the WMT'14 dataset, the translations produced by the LSTM achieve a BLEU score of 34.8 on the entire test set, where the LSTM's BLEU score was penalized on out-of-vocabulary words. Additionally, the LSTM did not have difficulty on long sentences. For comparison, a phrase-based SMT system achieves a BLEU score of 33.3 on the same dataset. When we used the LSTM to rerank the 1000 hypotheses produced by the aforementioned SMT system, its BLEU score increases to 36.5, which is close to the previous best result on this task. The LSTM also learned sensible phrase and sentence representations that are sensitive to word order and are relatively invariant to the active and the passive voice. Finally, we found that reversing the order of the words in all source sentences (but not target sentences) improved the LSTM's performance markedly, because doing so introduced many short term dependencies between the source and the target sentence which made the optimization problem easier.

Seq2Seq encoder-decoder paper, 2014

27. Kyunghyun Cho (b. ~1989)



Kyunghyun Cho, NYU

Key work / paper: *Learning Phrase Representations using RNN Encoder-Decoder...*, 2014 (GRU)

- Introduced the **Gated Recurrent Unit (GRU)** — simplified LSTM alternative
- Co-author (with Bahdanau, Bengio) of the 2014 attention-based NMT paper
- Encoder-decoder + attention framing is a direct conceptual precursor to the Transformer

arxiv.org/abs/1406.1078

Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation

Kyunghyun Cho

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Fethi Bougares Holger Schwenk

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Yoshua Bengio

Université de Montréal, CIFAR Senior Fellow

`find.me@on.the.web`

Abstract

In this paper, we propose a novel neural network model called RNN Encoder–Decoder that consists of two recurrent neural networks (RNN). One RNN encodes a sequence of symbols into a fixed-length vector representation, and the other decodes the representation into another sequence of symbols. The encoder and decoder of the proposed model are jointly

Along this line of research on using neural networks for SMT, this paper focuses on a novel neural network architecture that can be used as a part of the conventional phrase-based SMT system. The proposed neural network architecture, which we will refer to as an *RNN Encoder–Decoder*, consists of two recurrent neural networks (RNN) that act as an encoder and a decoder pair. The encoder maps a variable-length source sequence to a fixed-length vector, and the decoder maps the vec-

GRU / RNN Encoder-Decoder paper, 2014

Era 7: The Attention / Transformer Era

2017–2018

The architecture behind every modern LLM

28. Ashish Vaswani (b. ~1986)



Ashish Vaswani, formerly Google Brain

Key work / paper: *Attention Is All You Need*, 2017 (8 co-authors, Google)

- Introduced the **Transformer** — self-attention, no recurrence or convolution needed
- By consensus, the single most consequential AI paper of the last decade
- Underlies BERT, GPT (the “T” in GPT), and essentially every modern LLM (GPT-4, Claude, Gemini, LLaMA)

Wikipedia: en.wikipedia.org/wiki/Attention_Is_All_You_Need arxiv.org/abs/1706.03762

28. Ashish Vaswani (b. ~1986) — Key Work / Event

Attention Is All You Need



[Ashish Vaswani](#), [Noam Shazeer](#), [Niki Parmar](#), [Jakob Uszkoreit](#), [Llion Jones](#), [Aidan N. Gomez](#), [Lukasz Kaiser](#), [Illia Polosukhin](#)

[Ask](#)

powered by [CatalyzeX](#)

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

“Attention Is All You Need” Transformer paper, 2017

Era 8: The Modern Era

2016–now

Games, science breakthroughs, and the public AI boom

29. Demis Hassabis (b. 1976)



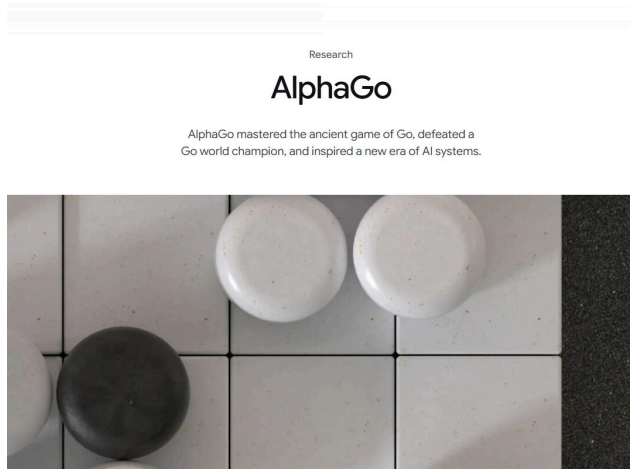
Demis Hassabis, CEO, Google DeepMind

Key work / paper: AlphaGo (2016); AlphaFold2 (2020)

- Co-founded DeepMind (2010); led **AlphaGo** — first program to beat a world-champion Go player
- Led **AlphaFold2** — solved the 50-year-old protein structure prediction problem
- **2024 Nobel Prize in Chemistry** (with John Jumper) — AI's first Nobel-recognized scientific breakthrough

Wikipedia: en.wikipedia.org/wiki/Demis_Hassabis

29. Demis Hassabis (b. 1976) — Key Work / Event



AlphaGo vs. Lee Sedol, 2016

30. Sam Altman (b. 1985)



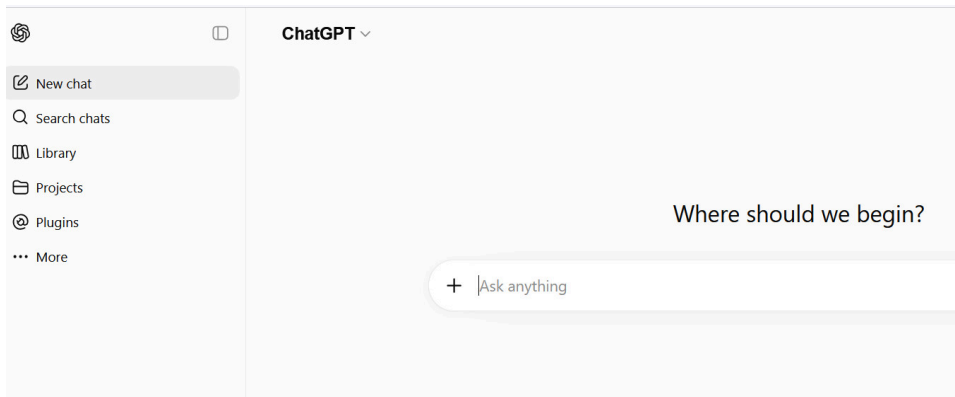
Sam Altman, CEO, OpenAI

Key work / paper: Supervised the launch of **ChatGPT**, November 30, 2022 (built on GPT-3.5)

- ChatGPT: 1 million users in 5 days, ~100 million monthly users within 2 months
- The moment AI became a mainstream global consumer phenomenon overnight
- 2023 ouster/reinstatement at OpenAI — landmark moment in the AI governance/safety debate

Wikipedia: en.wikipedia.org/wiki/Sam_Altman

30. Sam Altman (b. 1985) — Key Work / Event



ChatGPT launch announcement, Nov 30, 2022

Narrative Arc (1/6): Foundations & the Birth of AI

Era	Years	Theme	Key names
Foundations	1936–1955	Can machines think? Cybernetics, information theory	Turing, Wiener, Shannon
Birth of AI	1956–1969	Dartmouth Conference, symbolic AI, first neural nets, first chatbot	McCarthy, Minsky, Newell, Simon, Samuel, Rosenblatt, Weizenbaum

Narrative Arc (2/6): 1st AI Winter & Connectionist Revival

Era		Years	Theme	Key names
1st AI Winter		1969–1980	Perceptron critique kills neural net funding	(Minsky's <i>Perceptrons</i>)
Connectionist	Re-	1980–1998	Neuro-inspired vision, backprop, statistical/probabilistic learning	Fukushima, Hopfield, Rumelhart, Hinton, LeCun, Bengio, Vapnik, Pearl

Narrative Arc (3/6): 2nd AI Winter & Deep Learning Revolution

Era	Years	Theme	Key names
2nd AI Winter	late 1990s–2000s	Expert systems fail commercially; SVMs dominate	Vapnik era
Deep Learning Revolution	2006–2014	GPUs + big data + deep nets: ImageNet, AlexNet, GANs, seq2seq	Hinton, Fei-Fei Li, Krizhevsky, Sutskever, Goodfellow, Cho, Schmidhuber

Era	Years	Theme	Key names
RL & Games	2013–2017	Deep RL masters Atari, Go, StarCraft	Sutton, Barto, Hassabis

Narrative Arc (5/6): Attention / Transformer Era

Era	Years	Theme	Key names
Attention / Trans- former Era	2014–2018	Attention → Transformers → BERT/GPT	Bengio, Cho, Vaswani

Narrative Arc (6/6): Scale, Products & Alignment Era

Era	Years	Theme	Key names
Scale, Products & Alignment	2018–now	GPT-3/4, ChatGPT's public debut, RLHF, multimodal, science (AlphaFold), safety debates	Sutskever, Sutton/Barto, Hassabis, Hinton, Altman

- **Scaling laws & compute:** does scaling Transformers further yield AGI, or do we need new architectures?
- **Reasoning & agents:** models that plan, use tools, act autonomously (RL lineage + LLMs)
- **Multimodality:** one model for text, image, audio, video, robotics control
- **AI for science:** AlphaFold-style breakthroughs expanding to materials science, mathematics, drug discovery
- **Alignment & safety:** RLHF, constitutional AI, interpretability
- **Efficiency:** smaller, distilled, sparse (mixture-of-experts) models
- **Causality & world models:** Pearl-style causal reasoning to address hallucination

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Thank you!

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